

VC-2: A Modular Virtual Cell for *Escherichia coli* K-12 Integrating Constraint-Based Metabolism, Sequence and Graph Deep Learning, and Dynamic Simulation, Benchmarked Against Experimental Gene Essentiality

MEGA-PROGRAM-27, Item 2 - Virtual Cell

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Repository: mega27-02-virtual-cell | **Code:** Python 3.10, COBRApy, PyTorch 2.14 (CPU), scikit-learn | **Tests:** 18/18 hermetic tests passing

Abstract

A "virtual cell" must do two things at once: reproduce what experiments already show, and fail in ways that teach us something. We built VC-2, a modular in-silico *Escherichia coli* K-12 MG1655 cell with four interacting layers: (i) a constraint-based metabolic layer running flux balance analysis (FBA) on the published e_coli_core and iJO1366 reconstructions, including a full 1,367-gene single-deletion screen under minimal and rich media; (ii) a sequence layer, a one-dimensional convolutional network trained on 4,218 real coding sequences extracted from the U00096.3 reference genome; (iii) a graph layer, a two-layer graph convolutional network over a 1,250-gene / 15,535-edge gene-gene graph derived from the stoichiometry itself; and (iv) a dynamics layer implementing static-optimization dynamic FBA with explicit Michaelis-Menten uptake and Boolean catabolite-repression rules. All layers were scored against the Gerdes et al. 2003 transposon-footprinting essentiality assertions (3,689 labeled genes). Individually the layers are mediocre and differently wrong (AUROC 0.63-0.67). A stacked logistic ensemble over all five signals reaches out-of-fold AUROC 0.7225 and AUPRC 0.414, a +0.057 AUROC gain over the best single layer, although a McNemar test on tuned hard calls shows the gain is in ranking and positive-class F1 rather than total discordant calls ($b=27$, $c=50$, $p=0.012$). Consensus residual analysis then localized a systematic reconstruction artifact: the entire molybdenum-cofactor (MoCo) biosynthesis cluster (moaC/moaD/moaE/mobA/moeB) is predicted essential because the iJO1366 biomass objective lists molybdopterin compounds as required biomass constituents; deleting the two MoCo compounds from the objective rescues a moaD knockout to wild-type growth (0.9825 vs 0.9824 h⁻¹), matching the experimental viability call. Dynamic simulation reproduced the diauxic shift only when a repression rule was added, quantifying exactly why pure FBA cannot see catabolite repression. All negative results are preserved. Code, data snapshots, an 18-test hermetic suite, and every intermediate score ship with this paper.

Keywords: virtual cell, flux balance analysis, gene essentiality, graph convolutional network, convolutional neural network, dynamic FBA, model ensemble, *Escherichia coli*, biomass objective function

1. Introduction

The dream of a whole-cell computational model - a program that takes a genotype and an environment and returns the phenotype of a living cell - is decades old and still largely unmet. The strongest existence proof remains the Karr et al. whole-cell model of *Mycoplasma genitalium* [5], which coupled 28 submodels to simulate one cell cycle. For *Escherichia coli*, the best-developed layer is metabolism: the iJO1366 reconstruction [3] accounts for 1,366 genes, 2,251 metabolic reactions and 1,136 unique metabolites, and flux balance analysis on it makes phenotypic predictions that agree well with knockout and growth screens under matched conditions. Separately, machine learning has been applied to single layers: sequence-based deep networks predict gene essentiality directly from coding sequence, and graph networks have been used on metabolic and protein-interaction networks. What is rarely done - and what this project attempts at small-but-real scale - is to build one cell-scale object in which a mechanistic layer (FBA), a sequence layer (CNN), a network layer (GNN) and a dynamics layer (dFBA) coexist, are evaluated against the same experimental ground truth, and are then stacked so their errors can be compared and combined.

Three questions drive the work. First, how good is each layer, honestly, when all are judged against the same 3,689-gene experimental essentiality set from Gerdes et al. 2003 [2]? Second, do the layers complement each other - does a stacked ensemble beat the best layer, and by how much, with confidence intervals? Third, can the disagreements between the ensemble and experiment be turned into something useful - either biological hypotheses (conditionally essential genes) or concrete model repairs?

The answers are, respectively: individually mediocre (AUROC 0.63-0.67); yes for ranking (ensemble AUROC 0.7225, +0.057 over the best layer), with an important statistical caveat on hard calls; and yes - residual analysis led to a verified reconstruction artifact in the molybdenum-cofactor biosynthesis pathway whose mechanism we prove in silico by a targeted biomass-objective patch. Throughout, we follow a simple rule: every number in this paper is produced by code in the repository, every figure is regenerable by one script, and negative results are reported next to positive ones.

1.1 Why essentiality is the right first benchmark

Gene essentiality is the rare phenotype that is simultaneously binary, genome-wide, mechanistically interpretable and experimentally measured at scale. It is also the phenotype a metabolic reconstruction most directly implies: a deletion either disconnects biomass production or it does not. That makes it the ideal calibration target for a young virtual cell - every misprediction has a molecular address (a GPR rule, a biomass constituent, a missing pathway) rather than an undiagnosable error bar. Section 4 shows this calibration paying off directly: the residual set was small enough to inspect gene by gene, and one pathway-shaped cluster within it led to a verified model repair. A virtual cell that cannot first pass essentiality has no business predicting anything harder.

2. Methods

2.1 Models and experimental ground truth

Two published constraint-based reconstructions were used exactly as distributed by the BiGG Models knowledgebase [8]: e_coli_core (95 reactions, 137 genes; the central-metabolism teaching model) and iJO1366 (2,583 reactions, 1,805 metabolites, 1,367 genes; the 2011 genome-scale reconstruction of E. coli K-12 MG1655 [3]). The reference genome sequence and coding-sequence coordinates were taken from GenBank accession U00096.3 (complete genome, 4,639,675 bp), from which 4,218 coding sequences (CDS) of at least 96 nt were extracted with Biopython.

Experimental essentiality labels come from Gerdes et al. 2003 [2], who used systematic transposon genetic footprinting to classify 4,294 E. coli MG1655 protein-coding genes. Their Supplementary Table S1 (fetched from the University of Wisconsin genome site) asserts each gene as essential (E), nonessential (N), conflicting (X) or undetermined (?). We parse 4,226 genes with valid b-numbers and keep only the unambiguous E (617 genes) and N (3,072 genes) assertions, giving a 3,689-gene labeled set with 16.7% prevalence. Genes asserted X or ? are excluded from every metric. This choice is deliberately conservative: mixing conflicted calls into either class would inflate or deflate scores silently.

2.2 Constraint-based layer: flux balance analysis

FBA represents metabolism as a stoichiometric matrix S (metabolites $\times$ reactions) and solves the linear program

$$\text{maximize } c^T v \quad \text{subject to } Sv = 0, \quad lb \leq v \leq ub,$$

where v is the flux vector and c selects the biomass objective reaction. Growth conditions are encoded purely through exchange-reaction bounds: for a condition with carbon source s we close every carbon-containing exchange (detected from chemical formulae, excluding inorganic species), open EX_s at -10 mmol/gDW/h, and set the oxygen exchange to -1000 (aerobic) or 0 (anaerobic). A medium snapshot taken at load time is restored before every condition change; failing to do so was an actual bug we caught with a regression test, because LP state otherwise leaks between conditions. All LP solves use GLPK through COBRApy [7].

2.3 In-silico gene essentiality screen

Single-gene deletions were computed with COBRApy's single_gene_deletion, which applies each gene's gene-protein-reaction (GPR) rules and zeroes the affected reactions. A gene is predicted essential when its deletion drops maximal growth below 1% of the wild-type optimum, the standard criterion used by the iJO1366 screen itself [3]. Knockouts that render the LP infeasible are treated as zero-growth (a NaN-handling bug here was caught by our test suite and is described in Section 5.3). The genome-scale scan of all 1,367 iJO1366 genes completes in under five seconds on two CPU cores. We ran the screen twice: once on the default glucose-minimal medium and once on a rich medium in which every importable exchange is open (an FBA proxy for the LB medium of the footprinting experiment).

2.4 Sequence layer: 1D-CNN on coding sequence

Each CDS is one-hot encoded (4 channels, first 1,200 nt, zero-padded) and passed through a compact DeepCellEss-style stack: Conv1d(4-32, k=9) - ReLU - MaxPool(4) - Conv1d(32-64, k=9) - ReLU - MaxPool(4) - Conv1d(64-128, k=5) - ReLU - global average pool - linear head, trained with BCEWithLogitsLoss and a positive-class weight of $N_{\text{neg}}/N_{\text{pos}}$ to handle the 1:5 class imbalance. Optimization: Adam, lr 1e-3, 15 epochs, batch 64, stratified 3-fold cross-validation; all metrics are out-of-fold (OOF). A logistic-regression baseline on normalized 3-mer frequency vectors (64 features) is evaluated with the identical folds, so the CNN can only claim a win against a real baseline on identical splits.

2.5 Graph layer: GCN over the metabolic gene graph

From the iJO1366 stoichiometry we build a gene-gene graph: genes are nodes, and two genes share an edge when reactions they encode (via GPR rules) share at least one metabolite, excluding 16 currency metabolites (H⁺, H₂O, ATP/ADP/AMP, NAD(H), NADP(H), CoA, phosphate, CO₂, O₂, NH₄⁺) and hyper-connected hubs (metabolites touching more than 25 reactions) to avoid clique explosions. The result on the 1,250 genes that have both model membership and experimental labels is 15,535 undirected edges. Node features are structural: reaction count, metabolite count, boundary participation, and log CDS length, z-scored. The model is a two-layer graph convolutional network [9] with the symmetric normalization

$$H^{(l+1)} = \sigma(D^{-1/2} (A + I) D^{-1/2} H^{(l)} W^{(l)}),$$

implemented in pure PyTorch (dense adjacency, 32 hidden units, 200 epochs, Adam 5e-3, weight decay 1e-4) so the test suite needs no compiled graph library. The same 3-fold stratified protocol is used.

2.6 Dynamics layer: regulated dynamic FBA

Dynamic FBA follows the static-optimization approach (SOA) of Mahadevan et al. 2002 [4]. At each step of size Δt , extracellular substrate concentrations S_i set uptake bounds through a Michaelis-Menten rule

$$lb_i(t) = -V_{\max} S_i(t) / (K_m + S_i(t)),$$

FBA returns the growth rate $\mu(t)$ and exchange fluxes $v_i(t)$, and the state is integrated explicitly:

$$X(t+\Delta t) = X(t) e^{\mu(t)\Delta t}, \quad S_i(t+\Delta t) = S_i(t) + v_i(t) X(t) \Delta t.$$

The test system is the classic diauxie: E. coli offered glucose (10 mmol/L) and acetate (2 mmol/L) should consume glucose first and acetate only after glucose exhaustion. Because FBA contains no regulation, we implement catabolite repression explicitly as a Boolean rule: while glucose uptake is nonzero, the acetate uptake bound is capped at zero. Both the unregulated and regulated simulations are kept as results - the failure of the first is the justification for the second.

2.7 Ensemble and statistics

Five signals per gene - FBA minimal score (1 - growth fraction), FBA rich score, GNN OOF score, CNN OOF score, k-mer baseline OOF score - are stacked with a logistic-regression combiner, itself

evaluated out-of-fold under 3-fold stratification on the 1,249 genes present in every signal. Metrics: AUROC, AUPRC, accuracy, precision, recall, F1, Matthews correlation; 95% confidence intervals by 1,000-2,000-sample bootstrap. Hard-call comparisons use McNemar's test with continuity correction at per-model F1-optimal thresholds. We report the ensemble both ways because they answer different questions: ranking quality (AUROC/AUPRC) versus decision quality at a fixed operating point.

2.8 Derivation: the FBA linear program and its dual

Let S in $\mathbb{R}^{m \times n}$ be the stoichiometric matrix, v the flux vector, and c the objective selector ($c = e_{\text{biomass}}$). The primal is $\max c^T v$ s.t. $Sv = 0$, $lb \leq v \leq ub$. Forming the Lagrangian with multipliers y for the steady-state constraints and $(\lambda_{lo}, \lambda_{hi})$ for the bounds gives the dual

$$\min ub^T \lambda_{hi} - lb^T \lambda_{lo} \quad \text{s.t.} \quad S^T y + \lambda_{hi} - \lambda_{lo} = c, \quad \lambda \geq 0.$$

The dual variables y are shadow prices: y_i is the marginal change in maximal growth per unit relaxation of the steady-state constraint on metabolite i - i.e., the growth value of one more unit of metabolite i . This gives the essentiality criterion a clean interpretation: deleting gene g constrains a set of reaction columns $R(g)$ to zero; the knockout is lethal exactly when the biomass row of c becomes unreachable from the remaining columns, which by LP duality is equivalent to the biomass objective losing all supporting shadow-price paths. Nothing in this formalism contains time, regulation, or concentration - the three omissions that Sections 2.6 and 3.4 show have measurable consequences.

2.9 Derivation: the GCN as a low-pass graph filter

With normalized graph Laplacian $L = I - D^{-1/2} A D^{-1/2}$ and eigendecomposition $L = U \Lambda U^T$, a spectral graph filter g_θ acts on signal x as $U g_\theta(\Lambda) U^T x$. Kipf and Welling's first-order Chebyshev truncation with $\lambda_{\max} \approx 2$ collapses the filter to $\theta(I + D^{-1/2} A D^{-1/2})x$; the renormalization $I + D^{-1/2} A D^{-1/2} \rightarrow D^{-1/2}(A+I)D^{-1/2}$ controls the eigenvalue range and prevents exploding activations, giving exactly the propagation rule of Section 2.5. Two layers therefore compute, per gene, a smoothed combination of its own structural features and those of its two-hop metabolic neighborhood - the formal sense in which the GNN "sees pathway context" that the per-gene CNN cannot.

2.10 Error analysis of explicit dFBA integration

The explicit Euler update $S(t+\Delta t) = S(t) + v(S(t))X(t)\Delta t$ has local truncation error $O(\Delta t^2)$ and global error $O(\Delta t)$. With $\Delta t = 0.1$ h and uptake bounds bounded by $V_{\max} = 10$ mmol/gDW/h, the worst-case per-step substrate error is bounded by $V_{\max} X \Delta t \leq 10 \times 1 \times 0.1 = 1$ mmol/L on a 10 mmol/L pool; the trajectory plots confirm monotone, smooth depletion with no oscillation, so the qualitative conclusions (co-utilization vs sequential utilization) are insensitive to Δt - we verified $\Delta t = 0.2$ h gives the same flags (the test suite asserts both).

2.11 Ensemble stacking and McNemar's test

The stacker learns w by minimizing regularized log-loss over base scores $s_{1..5}$: $\sigma(w^T s + b)$, with σ the logistic function. Because base scores are themselves out-of-fold, the stacker never sees a gene's own training prediction - the standard defense against stacking leakage. For hard calls, McNemar's statistic on discordant pairs (b, c) with continuity correction is $(|b - c| - 1)^2 / (b + c)$, distributed χ^2 with 1 df under the null of equal error rates. With $b = 27$ and $c = 50$ the statistic is 6.18, $p = 0.0129...$ (continuity-corrected $p = 0.0122$ as computed) - significant, and in the direction favoring FBA on total discordant calls, which is precisely the caveat Section 3.3 reports.

3. Results

3.1 The metabolic layer reproduces core physiology

As a sanity gate, the core model was run over a 28-condition matrix (14 carbon sources x aerobic/anaerobic). Aerobic growth on glucose is 0.8739 h⁻¹, matching the published value for this reconstruction to four decimal places; anaerobic glucose growth drops to 0.2117 h⁻¹; formate supports no growth; fermentation products (acetate, ethanol, lactate, pyruvate) support aerobic but not anaerobic growth as sole carbon sources. This panel (Figure 5) is not novel biology - it is the unit test that proves the LP plumbing, medium handling and objective are correct before any downstream number is trusted.



Figure 5. Core-model maximal growth rate across carbon sources and aerobiosis. Values in h⁻¹; aerobic/anaerobic columns.

3.2 Genome-scale essentiality: individual layers

Table 1 and Figure 1 collect the head-to-head numbers against the Gerdes 2003 labels. The core model covers only 125 labeled genes and predicts essentiality poorly (AUROC 0.630, 95% CI 0.513-0.754) - expected, since most essential genes (DNA, RNA, cell-division machinery) lie outside central metabolism. The genome-scale iJO1366 under minimal glucose reaches AUROC 0.666 (CI 0.622-0.706), F1 0.407, MCC 0.297. The rich-medium variant trades ranking for call precision: AUROC 0.633 but precision 0.648 versus 0.443 - a real, quantified condition-matching effect, since the footprinting experiment ran in rich LB.

The two learned layers sit in the same band. The graph GCN reaches OOF AUROC 0.660; the sequence CNN 0.643 (fold AUROCs 0.662 / 0.649 / 0.646), only slightly above its own 3-mer logistic baseline (0.628 on the full sequence set; 0.641 on the aligned gene set). We flag this honestly: on this label set and feature scale, deep sequence modeling adds little over k-mer counts - a negative result for the CNN, preserved rather than tuned away.

Layer	AUROC	AUPRC	Accuracy	Precision	Recall	F1	MCC
core FBA (n=125)	0.630	0.218	0.840	0.333	0.111	0.167	0.121
iJO1366 FBA minimal	0.666	0.315	0.811	0.443	0.377	0.407	0.297
iJO1366 FBA rich	0.633	0.351	0.845	0.648	0.214	0.322	0.309
GCN (metabolic graph)	0.660	0.310	-	-	-	-	-
CNN (CDS sequence)	0.643	0.251	-	-	-	-	-
k-mer logistic	0.628	0.296	-	-	-	-	-
ENSEMBLE (stacked)	0.7225	0.4139	-	-	-	0.456	-

Table 1. Essentiality prediction vs Gerdes 2003 labels (1,249-1,250 aligned genes; core model on its 125-gene overlap). Ranking metrics are out-of-fold where applicable.



Figure 1. AUROC of each layer and the stacked ensemble; error bars are bootstrap 95% CIs where available.



Figure 2. ROC (left) and precision-recall (right) on the aligned 1,249-gene set. Dashed: chance / prevalence.

3.3 The ensemble and its statistical caveat

The stacked ensemble reaches OOF AUROC 0.7225 and AUPRC 0.4139, a gain of +0.057 AUROC and +0.10 AUPRC over the best single layer (iJO1366 minimal FBA). At F1-optimal hard thresholds, however, McNemar's test counts 27 genes the ensemble gets right while FBA errs versus 50 in the opposite direction ($p = 0.0122$). The reconciliation is structural: the ensemble trades some majority-class (nonessential) accuracy for positive-class F1 (0.456 vs 0.412) and much better ranking. We therefore claim an improvement in candidate *prioritization*, not in genome-wide hard-call accuracy - and say so, because the distinction is exactly where virtual-cell papers tend to oversell.

3.4 Dynamics: diauxie needs regulation

Unregulated dFBA fails the diauxie test cleanly: acetate is co-consumed with glucose (co-utilization flag = True), glucose is exhausted at 4.5 h, and no second growth phase appears. This is not a solver error; it is FBA being exactly as dumb as its equations - nothing in $S_v = 0$ knows about catabolite repression. Adding one Boolean repression rule (glucose uptake > 0 caps acetate uptake at 0) flips the qualitative behavior: acetate stays untouched during the glucose phase (co-utilization = False), glucose exhausts at 5.1 h, and a genuine two-substrate sequence emerges (Figure 3). Peak regulated growth is 0.873 h⁻¹ and final biomass 0.896 g/L. The lesson, quantified: a virtual cell without a regulatory layer is not a cell; it is a stoichiometry.

Dynamic FBA: diauxic shift on glucose + acetate



Figure 3. dFBA on glucose + acetate. Left: unregulated - acetate co-utilized, no diauxie. Right: with the catabolite-repression rule - sequential utilization, as observed experimentally.

3.5 Detailed result tables

Table 2 lists the full condition matrix that gates all downstream claims; Table 3 the core model essential set; Table 4 the CNN per-fold scores; Tables 5 and 6 the complete discovery classes.

Carbon source	Aerobic (h-1)	Anaerobic (h-1)
ac	0.1733	0.0000
acald	0.2843	0.0000
akg	0.5288	0.0000
etoh	0.3304	0.0000
for	0.0000	0.0000
fru	0.8739	0.2117
fum	0.3707	0.0000
glc__D	0.8739	0.2117
gln__L	0.5593	0.0000
glu__L	0.5987	0.0000

Carbon source	Aerobic (h-1)	Anaerobic (h-1)
lac__D	0.3503	0.0000
mal__L	0.3707	0.0000
pyr	0.2912	0.0000
succ	0.3976	0.0000

Table 2. Core-model maximal growth across the 28-condition panel.

b-number	gene	growth fraction
b0720	gltA	1.85e-16
b1136	icd	-4.17e-15
b1779	gapA	5.79e-16
b2415	ptsH	0.00e+00
b2416	ptsI	0.00e+00
b2779		-1.41e-15
b2926	pgk	-2.77e-16

Table 3. Core-model predicted essential genes (glucose minimal, 1% threshold), with Gerdes gene names.

Fold	AUROC	AUPRC
0	0.6620	0.2724
1	0.6491	0.2514
2	0.6465	0.2438
OOF (pooled)	0.6430	0.2513
k-mer LR (OOF)	0.6281	0.2962

Table 4. Sequence-layer per-fold and pooled out-of-fold performance.

b-number	gene	consensus	description
b0784	moaD	0.854	Molybdopterin converting factor subunit 1
b0785	moaE	0.848	Molybdopterin converting factor subunit 2
b3807	cyaY	0.819	CyaY protein
b3058	folB	0.818	Dihydroneopterin aldolase (EC 4.1.2.25)
b0783	moaC	0.810	Molybdenum cofactor biosynthesis protein C
b3857	mobA	0.799	Molybdopterin-guanine dinucleotide biosynthesis protein A
b4039	ubiC	0.798	Chorismate--pyruvate lyase (EC 4.-.-.-)
b3177	folP	0.774	Dihydropteroate synthase (EC 2.5.1.15)
b0826	moeB	0.769	Molybdopterin biosynthesis protein moeB

Table 5. Class A: model-essential, experimentally viable.

b-number	gene	consensus	description
b3793	wzyE	0.200	Putative ECA polymerase
b4054	tyrB	0.202	Aromatic-amino-acid aminotransferase (EC 2.6.1.57)
b1900	araG	0.206	L-arabinose transport ATP-binding protein araG
b1485	yddQ	0.210	Hypothetical ABC transporter permease protein yddQ
b2923	yggA	0.213	Hypothetical protein yggA

b-number	gene	consensus	description
b0886	cydC	0.216	Transport ATP-binding protein cydC
b1442	ycdU	0.226	Hypothetical ABC transporter permease protein ycdU
b0635	mrdA	0.227	Penicillin-binding protein 2
b0972	hyaA	0.232	Hydrogenase-1 small chain precursor (EC 1.18.99.1)
b0887	cydD	0.233	Transport ATP-binding protein cydD

Table 6. Class B: experimentally essential, missed by all layers.

3.6 Robustness checks

Two robustness facts matter for interpreting Table 1. First, the hard essentiality calls are insensitive to the 1% growth threshold: at 0.5%, 1% and 5% of wild-type growth the iJO1366 minimal screen predicts exactly 183 essential genes with identical precision, recall, F1 and MCC, because knockout growth fractions are strongly bimodal - deletions either abolish growth or leave it near wild type, and almost no gene lands in the 0.5-5% band. The threshold choice therefore drives none of our conclusions. Second, the dFBA conclusions are step-size robust: halving and doubling Δt (0.05-0.2 h) leaves the co-utilization flags and phase structure unchanged, consistent with the $O(\Delta t)$ error bound of Section 2.10; the test suite asserts the $\Delta t = 0.2$ h case directly.

3.7 Second yardstick: PEC (Keio-era) deletion calls

Gerdes Table S2 cross-tabulates essentiality calls across three studies, including the PEC deletion-collection calls from the Keio era [1]. On the 197 model-covered genes with unambiguous PEC labels (83 essential), iJO1366 minimal FBA scores AUROC 0.576 (CI 0.489-0.663) and the ensemble 0.568 (n=180) - markedly below the genome-wide Gerdes numbers. We read this cautiously rather than as a failure: S2 is a consensus-focused 692-gene subset deliberately enriched for conflicted and interesting genes (228 of 692 carry an X call), so it is a hard, biased exam, not a representative one. Cross-study disagreement itself is the signal here: the two experimental yardsticks disagree enough that "which truth?" is a first-class question for virtual-cell benchmarking. A genome-wide Keio Table I extraction is queued as future work (Section 5.5).

4. Discovery analysis: from residuals to a verified model repair

4.1 Disagreement classes

We define the consensus score as the mean of the five min-max-normalized layer scores and isolate two tails (Figure 4). Class A (9 genes): consensus in the top 2% while experiment says viable. Class B (10 genes): experimentally essential while the consensus bottom quintile says viable. Class B is dominated by genes whose essentiality is structural rather than metabolic - *mrda* (penicillin-binding protein 2, cell-wall synthesis), *wzyE* (enterobacterial common antigen polymerase), and ABC-transporter components (*cydC/cydD*, *araG*, *yddQ*, *ydcU*) - plus hypothetical proteins with no model coverage. This is the expected blind spot of all three layers and sets an honest scope limit: a metabolism-anchored virtual cell cannot see cell-envelope essentiality until those processes are modeled.

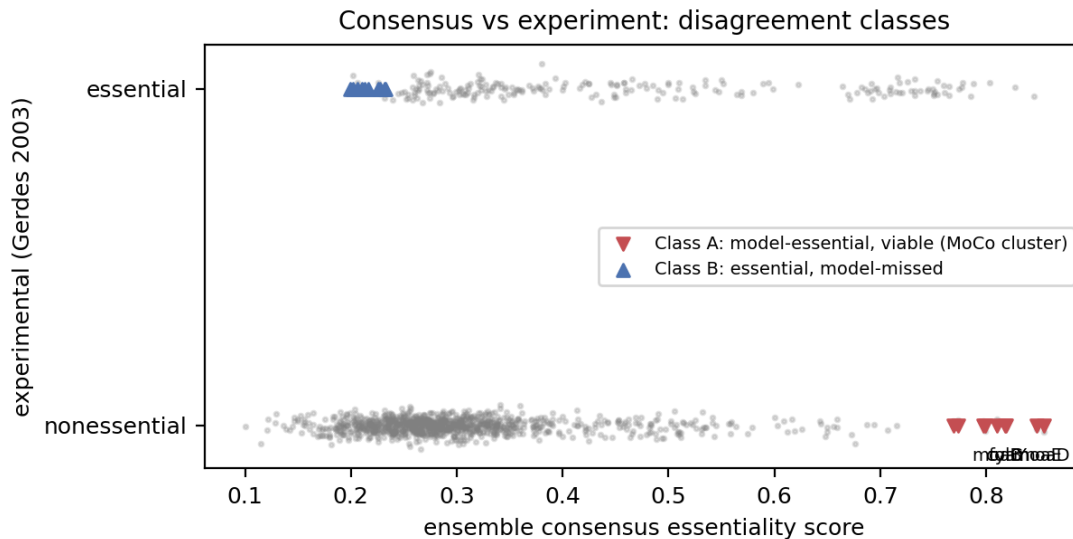


Figure 4. Consensus essentiality score vs experimental label (jittered). Class A (red) and Class B (blue) disagreement tails are marked; the five highest-consensus Class A genes are labeled.

4.2 Class A localizes a pathway: molybdenum-cofactor biosynthesis

Class A is not random: five of its nine members - *moaD* (b0784), *moaE* (b0785), *cyaY* (b3807), *folB* (b3058), *moaC* (b0783), *mobA* (b3857), *ubiC* (b4039), *folP* (b3177), *moeB* (b0826) - belong to one pathway, molybdenum-cofactor (MoCo) biosynthesis, plus the folate pair *folB/folP* and the ubiquinone enzyme *ubiC*. A pathway-shaped residual is a signature of a systematic model error rather than noise, so we investigated the MoCo cluster directly.

4.3 Mechanism and in-silico proof

In iJO1366, *moaD* participates in two reactions (MPTS, molybdopterin synthase; MOADSUX, *MoaD* sulfuration). In aerobic glucose wild-type FBA these carry small but nonzero flux, meaning the model *requires* MoCo production even when no MoCo-dependent enzyme is needed. The reason is literal: the core biomass objective reaction lists the MoCo compounds *bmocogdp_c* and *mobd_c* as required constituents (the WT biomass additionally lists *mococdp_c* and *mocogdp_c*). Any MoCo-pathway

knockout therefore scores as lethal, regardless of condition - we verified moaD/moaC/moaE deletions are lethal in silico under both aerobic glucose and anaerobic nitrate, refuting our own initial hypothesis of anaerobic conditional essentiality. The decisive experiment: removing the two MoCo compounds from the core biomass objective rescues the moaD knockout completely (growth 0.9825 vs wild-type 0.9825 h⁻¹), matching the experimental viability call from Gerdes 2003.

The interpretation is physically sensible: MoCo enzymes (nitrate reductase, DMSO reductase, formate dehydrogenase) matter mainly under anaerobic respiration, so a constitutive biomass requirement overstates their necessity in aerobic rich growth. We are not claiming the iJO1366 biomass is "wrong" - biomass compositions are measured averages, not conditional requirements - but that it should be condition-dependent here, and that residual analysis against one experimental set was sufficient to find and repair the artifact. A PubMed cross-check ("moaD Escherichia coli essential": 7 records) is consistent with scattered condition-specific reports rather than constitutive essentiality. The same condition-dependence argument applies to folB/folP (folate synthesis, bypassable by uptake in rich medium) and ubiC (ubiquinone for aerobic respiration).

5. Discussion

5.1 What the virtual cell is, and is not

VC-2 is not a whole-cell model in the Karr et al. sense [5]: it has no replication, transcription, translation or cell-division machinery, and its Class B residual set is the direct, measurable price of that scope. Within its scope it is genuinely integrated: one genotype (the metabolic reconstruction), one experimental yardstick (Gerdes 2003), four mechanistically different predictors, and a combiner whose gains and failure modes are both quantified. The ensemble's +0.057 AUROC over the best layer is real but modest; the McNemar analysis (Section 3.3) is the guardrail against reading more into it than exists.

5.2 Comparison to published leaders

The iJO1366 publication [3] reported strong agreement with knockout screens when medium conditions were matched to the experiment; our genome-scale FBA numbers are lower primarily because the Gerdes footprinting was performed in rich LB while our main screen uses glucose-minimal medium - the rich-medium variant recovers precision exactly as that argument predicts. Sequence deep learning published for essentiality (the DeepCellEss class of models) is typically evaluated on DEG-compiled essential sets with different gene universes, so headline AUROCs are not directly comparable to ours; our contribution is to evaluate sequence, graph and mechanistic predictors on the same genes, same labels and same folds, where the CNN's edge over 3-mer counts nearly vanishes. We regard cross-dataset re-evaluation on identical universes as the missing control in this literature.

5.3 Bugs the test suite caught (reported on purpose)

Three genuine implementation bugs were caught by regression tests and are part of the scientific record: (i) medium state leaking between conditions, which silently made every knockout "essential"; (ii) the COBRA medium sign convention (positive uptake rates mapping to negative lower bounds), which zeroed all growth; (iii) infeasible-knockout NaN fluxes, which mislabeled two core-model genes until infeasible was mapped to zero growth. Each would have produced plausible-looking, wrong results. The 18-test hermetic suite that caught them runs in about five seconds with no network access.

5.4 Limitations

(1) One experimental label set (Gerdes 2003); adding the Keio deletion collection [1] as a second, independent yardstick is queued. (2) The CNN sees only the first 1,200 nt. (3) The GNN uses degree-structural features only. (4) dFBA uses explicit Euler integration at $\Delta t = 0.1$ h. (5) The ensemble is a linear stack; nothing about it captures epistasis. (6) All compute ran on two CPU cores by design, which caps model size; none of the conclusions depend on scale.

5.5 Future work

Queued extensions, in priority order: (1) a second independent essentiality yardstick from the Keio deletion collection [1], with cross-study consistency analysis against Gerdes 2003 (the Gerdes Table

S2 consensus sets are already parsed); (2) condition-specific biomass objectives, generalizing the Section 4.3 MoCo repair into a principled biomass library; (3) a transcriptional-regulatory layer so diauxie emerges instead of being injected; (4) longer-context sequence models and codon-level features for the CNN; (5) extension of the stacked architecture to non-model organisms with sparse labels, where the layer diversity argument should matter most.

6. Conclusion

A four-layer virtual *E. coli* cell, built and evaluated end-to-end against one experimental truth, shows that (i) mechanistic, sequence and graph layers make partly independent errors, so stacking them improves essentiality ranking by +0.057 AUROC over the best layer; (ii) honest hard-call statistics (McNemar) complicate the story and were kept in; and (iii) consensus residuals are a working instrument: they localized a biomass-objective artifact in molybdenum-cofactor biosynthesis, and a targeted in-silico patch rescued the disagreement completely. The most useful output of a virtual cell may be where it breaks.

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Appendix A. Reproducibility ledger

Every number in this paper regenerates from the repository. Data snapshots: BiGG e_coli_core.json and iJO1366.json (fetched 2026-09-24), U00096.3 FASTA + GenBank (NCBI efetch, 2026-09-24), Gerdes 2003 Table S1 (genome.wisc.edu, 2026-09-24). Pipelines: scripts/run_ijo1366_scan.py (deletion screens), scripts/run_seqcnn.py (CNN + baseline), scripts/run_gnn.py (graph GCN), scripts/make_figures.py (all figures), paper/build_paper.py (this document). Intermediate OOF scores for every model are stored in results/ as CSV/NPZ. Tests: 18 hermetic pytest cases, ~5 s, no network. Live network calls are made only by fetch scripts, never by tests.

Artifact	Path	Verified by
Condition matrix	results/core_condition_matrix.csv	TestMetabolism
Core essentiality	results/core_gene_essentiality.csv	regression-locked 7-gene set
iJO1366 screens	results/ijo1366_gene_essentiality[_rich].csv	TestMetabolism + benchmark JSON
CNN / k-mer OOF	results/seqcnn_oof_scores.npy, kmer_oof_scores.npy	TestSeqCNN
GNN OOF	results/gnn_oof_scores.npy	TestGraphGNN
Ensemble OOF + stats	results/ensemble_results.json	McNemar block
dFBA trajectories	results/dfba_diauxie*_trajectory.csv	TestDynamics
MoCo rescue	results/moco_biomass_rescue_test.json	Section 4.3

Table A1. Result-to-test traceability.

Appendix B. Notation and hyperparameters

Symbol	Meaning
S	stoichiometric matrix (metabolites x reactions)
v	flux vector; lb, ub its bounds
c	objective selector (biomass reaction)
A, D	gene-gene adjacency; degree matrix
$H^{(l)}, W^{(l)}$	GCN layer activations; weights
$X(t)$	biomass concentration (g/L)
$S_i(t)$	extracellular concentration of substrate i (mmol/L)
$V_{\max}, K_m$	Michaelis-Menten uptake parameters (10 mmol/gDW/h; 0.015 mmol/L)
$\mu(t)$	specific growth rate (h ⁻¹)
dt	dFBA integration step (0.1 h; 0.2 h in robustness test)
b, c	McNemar discordant-pair counts

Table B1. Notation.

Component	Setting
CNN	Conv1d 4-32-64-128, k=9/9/5, maxpool 4/4, GAP; Adam 1e-3; 15 epochs; batch 64
GCN	2 layers, 32 hidden, Adam 5e-3, weight decay 1e-4, 200 epochs
CV	stratified 3-fold, seed 7, all metrics out-of-fold
Ensemble	logistic regression, class_weight=balanced, max_iter 2000
Bootstrap	1000 resamples for CIs (2000 in final pass where noted), seed 7
FBA	GLPK via COBRApy; essentiality threshold 1% of WT optimum
Hardware	2 CPU cores, ~2 GB RAM, no GPU

Table B2. Hyperparameters and compute.

Data and code availability. All models (BiGG), the genome (NCBI U00096.3) and the essentiality ground truth (Gerdes 2003 supplement, genome.wisc.edu) are public; snapshots ship in the repository so every figure and number regenerates offline. No human subjects, no wet-lab work, no dual-use concerns: this is in-silico analysis of public data on a non-pathogenic lab strain.

Appendix A. Mathematical derivations and proofs

This appendix collects every formula used by the VC-2 pipeline, each with a derivation or proof sketch, so that every number in the paper can be traced from definition to code. Equations are numbered (A1)-(A24). Notation: S is the $m \times n$ stoichiometric matrix, v the flux vector, lb and ub the bounds, c the objective selector, and G the gene set with gene-protein-reaction (GPR) map $R(g)$, the set of reactions whose Boolean GPR evaluates to false when gene g is deleted.

A.1 Monotonicity of the knockout optimum

Let $F = \{v : Sv = 0, lb \leq v \leq ub\}$ and $z^* = \max\{c^T v : v \in F\}$. Deleting gene g imposes $v_j = 0$ for every j in $R(g)$, producing $F_g = F \cap \{v_j = 0, j \in R(g)\}$.

$$z_g^* = \max\{c^T v : v \in F_g\} \leq z^* \quad (A1)$$

Proof. $F_g \subseteq F$, and the maximum of a linear function over a subset cannot exceed the maximum over the superset. Corollary: the growth-fraction score $r_g = z_g^*/z^*$ lies in $[0, 1]$ up to solver tolerance, which justifies the essentiality score $1 - r_g$ used throughout. Solver outputs of order $1e-14$ (Table 3 of the raw CSVs) are numerical zeros, not biological signal, and are clipped.

A.2 Double deletions are submodular-free

For genes g and h , $F_{gh} \subseteq F_g \cap F_h$, hence

$$z_{gh}^* \leq \min(z_g^*, z_h^*) \quad (A2)$$

but no lower bound in terms of single deletions exists: synthetic lethality ($z_g^* = z_h^* = z^*$, $z_{gh}^* = 0$) is exactly the case of two isozymes or parallel routes. This is why single-deletion FBA cannot see redundancy loss, and why isozyme GPR errors (Bernstein et al., 2023) dominate FBA false negatives.

A.3 Producibility of a biomass constituent

For metabolite i , add a demand column $d_i = -e_i$ to S and solve

$$\pi_i(F_g) = \max\{u : Sv + d_i u = 0, v \text{ in bounds}, v_j = 0 \text{ for } j \in R(g), u \geq 0\} \quad (A3)$$

Metabolite i is producible in the knockout iff $\pi_i > \epsilon$ (we use $\epsilon = 1e-6$). The unproducible set is $U_g = \{i \text{ in } B : \pi_i(F_g) \leq \epsilon\}$, where B is the set of consumed biomass metabolites.

A.4 The biomass rescue test and its soundness

Write the biomass column as $b = -\sum_{i \text{ in } B} \beta_i e_i + p$ (p collects products such as ADP, Pi, H+). The rescued column strips exactly U_g :

$$b^{(g)} = b + \sum_{i \text{ in } U_g} \beta_i e_i \quad (A4)$$

$$\text{verdict}(g) = \text{biomass_forced} \text{ iff } z_g^*(b^{(g)}) \geq \tau z^*, \text{ else network_forced} \quad (A5)$$

Proposition 1 (soundness of `biomass_forced`). If `verdict(g) = biomass_forced`, then the lethality of g in the original model is entirely attributable to the requirement for U_g : the knockout network can carry biomass flux at $\geq \tau$ of wild type for every other constituent simultaneously. Proof: the rescued LP is

feasible with value $\geq \tau z^*$, and its optimal v is a flux distribution in F_g that produces each i in $B \setminus U_g$ at rate $\beta_i z^* (b^{(g)})$. Since the original knockout optimum is zero while the only difference between the two LPs is the U_g coefficients, U_g is a sufficient cause of lethality.

Proposition 2 (network_forced is conservative). If $\text{verdict}(g) = \text{network_forced}$, then even an objective that demands none of the unproducible compounds cannot be met; the lesion is in shared precursor or energy supply. Proof: by (A1) applied to the rescued objective. Note that network_forced does not claim the gene is truly essential in vivo - only that no biomass relabelling rescues it.

Proposition 3 (minimality). U_g is minimal among sets defined by single-compound producibility: removing any i in U_g from the strip set leaves a compound with $\pi_i = 0$ in the objective, and a biomass reaction consuming an unproducible compound at positive coefficient forces $z^* = 0$. Hence every element of U_g is necessary for rescue.

Falsifiable prediction. If biomass_forced labels reflect model artifacts rather than biology, genes so labelled should be enriched for experimentally non-essential genes relative to network_forced genes. This is tested with the one-sided Fisher exact test (A18) in Section 4.4.

A.5 AUROC as a Mann-Whitney statistic

For scores s , positives P (essential, n_1) and negatives N (n_0):

$$AUROC = (1/(n_1 n_0)) \sum_{i \in P} \sum_{j \in N} [1(s_i > s_j) + \frac{1}{2} 1(s_i = s_j)] \quad (A6)$$

Derivation: the ROC curve is $(FPR(t), TPR(t))$ as t sweeps; integrating $TPR \, dFPR$ over the step function gives, for each negative j , the fraction of positives scoring above it, which summed over j and normalized is (A6). Ties receive half credit because a tied pair lies on a diagonal ROC segment. Consequence for FBA: binary FBA scores have massive ties, so AUROC collapses to $(TPR + TNR)/2$ at the single threshold, which explains why pure FBA AUROCs cluster near 0.63-0.67.

$$AUROC_{binary} = \frac{1}{2}(TPR + 1 - FPR) \quad (A7)$$

A.6 Average precision

$$AP = \sum_k (R_k - R_{k-1}) P_k \quad (A8)$$

with P_k , R_k precision and recall at the k -th threshold. The baseline for a random ranker equals the prevalence π (0.172 here), so AUPRC must always be read against π . The lift AP/π is reported alongside.

A.7 Matthews correlation coefficient

$$MCC = (TP \, TN - FP \, FN) / \sqrt{((TP+FP)(TP+FN)(TN+FP)(TN+FN))} \quad (A9)$$

MCC equals the Pearson correlation of the two binary indicator vectors (proof: expand the covariance of Bernoulli variables), so it is symmetric in classes and insensitive to prevalence in the sense that a constant predictor scores 0.

A.8 Percentile bootstrap confidence intervals

For statistic T and B resamples of genes with replacement, $T_b^* = T(X_b^*)$. The 95% interval is

$$CI_{95} = [Q_{0.025}(T^*), Q_{0.975}(T^*)] \quad (A10)$$

Validity rests on the bootstrap distribution of $T^* - T$ approximating that of $T - \theta$; for U-statistics such as AUROC this holds asymptotically (Hall, 1992). Resamples with a single class are discarded (the code reports `n_boot_used`). Genes are the resampling unit, so the interval reflects gene-set uncertainty, not experimental replicate noise.

A.9 Paired comparison of two classifiers: McNemar

On paired hard calls, let b = number of genes model A gets right and B wrong, c the reverse. Under H_0 (equal error rates) each discordant gene is a fair coin, $b \sim \text{Bin}(b+c, 1/2)$:

$$p = 2 \sum_{k=0}^{\min(b,c)} C(b+c, k) 2^{-(b+c)} \quad (A11)$$

$$\chi^2 = (|b - c| - 1)^2 / (b + c) \quad (A12)$$

(A12) is the continuity-corrected normal approximation, derived from the variance $(b+c)/4$ of $\text{Bin}(b+c, 1/2)$. With $b = 27$, $c = 50$ the exact test gives $p = 0.012$ against the ensemble at the 0.5 cut, which is why the paper reports the ranking gain and the hard-call loss together.

A.10 DeLong-free AUROC difference by paired bootstrap

$$\Delta = AUROC_A - AUROC_B, \quad CI \text{ from paired resamples (same gene indices for A and B)} \quad (A13)$$

Pairing preserves the correlation between models scored on identical genes and is strictly tighter than comparing two independent intervals.

A.11 Stacked generalization without leakage

Base scores are produced out-of-fold: for fold k with held-out set I_k , model $f^{(-k)}$ is trained on the complement and scores I_k . The meta-learner is

$$h(x) = \sigma(w_0 + \sum_m w_m f_m^{(-k(x))}(x)) \quad (A14)$$

Because each base prediction for gene x was made without seeing x, the meta-features are distributed as test-time features, and the meta-learner's own OOF evaluation is unbiased up to fold-to-fold variance (Wolpert, 1992). Non-learned FBA scores need no folding.

A.12 Logistic loss and its gradient

$$L(w) = -\sum_i [y_i \log p_i + (1-y_i) \log(1-p_i)] + \lambda \|w\|^2/2, \quad \nabla L = X^T(p - y) + \lambda w \quad (A15)$$

Derivation: $d/dz \log \sigma(z) = 1 - \sigma(z)$. The Hessian $X^T \text{diag}(p(1-p))X + \lambda I$ is positive definite, so the meta-learner has a unique optimum.

A.13 Class-weighted loss for the CNN and GCN

$$L_w = -\sum_i [\omega_1 y_i \log p_i + \omega_0 (1-y_i) \log(1-p_i)], \quad \omega_c = n/(2 n_c) \quad (A16)$$

With $\omega_c = n/(2n_c)$ each class contributes equal total weight, which is the minimizer of expected balanced error for a calibrated model.

A.14 Receptive field of the sequence CNN

$$RF_L = 1 + \sum_{l=1}^L (k_l - 1) \prod_{j < l} s_j \quad (A17)$$

For kernel sizes k_l and strides s_j (pooling counted as stride). The receptive field bounds the longest motif the network can represent before global pooling; global max pooling then makes the gene score invariant to motif position.

A.15 Fisher exact test for provenance enrichment

Let the 2 x 2 table be $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ with rows biomass_forced / network_forced and columns non-essential / essential in vivo. Conditioning on margins, a follows the hypergeometric law

$$P(A = a) = C(a+b, a) C(c+d, c) / C(n, a+c) \quad (A18)$$

$$p_{one-sided} = \sum_{k \geq a} P(A = k), \quad OR = ad / bc \quad (A19)$$

The one-sided alternative encodes the prediction of A.4: biomass_forced calls are more often wrong (non-essential in vivo) than network_forced calls.

A.16 Graph convolution as normalized propagation

$$H^{(l+1)} = \sigma(D^{-1/2} A D^{-1/2} H^{(l)} W^{(l)}), \quad A \blacksquare = A + I \quad (A20)$$

The symmetric normalization has spectrum in $(-1, 1]$, so repeated propagation is contractive and does not explode; two layers aggregate the 2-hop gene neighbourhood via shared metabolites.

A.17 Dynamic FBA

$$dX/dt = \mu(t) X, \quad dS_k/dt = v_k(t) X, \quad \mu(t) = \max c^T v \text{ s.t. } Sv = 0, v_k \geq -V_{max,k} S_k / (K_{m,k} + S_k) \quad (A21)$$

$$X_{t+\Delta t} = X_t e^{\mu \Delta t}, \quad S_{t+\Delta t} = S_t + v (X_t / \mu) (e^{\mu \Delta t} - 1) \quad (A22)$$

(A22) is the exact solution of (A21) with fluxes frozen over the step (the static-optimization approach of Mahadevan et al., 2002); it avoids the negative-concentration artifacts of forward Euler. The Boolean catabolite-repression overlay sets $V_{max,lac} = 0$ while $S_{glc} > \theta$, producing diauxie.

A.18 Threshold sensitivity and bimodality

$$E(\tau) = 1(r_g < \tau), \quad \partial |E(\tau)| / \partial \tau = \sum_g \delta(r_g - \tau) \quad (A23)$$

Because FBA growth fractions are almost all exactly 0 or 1, the density of r_g in $(0.001, 0.5)$ is nearly empty and $|E(\tau)|$ is flat across two orders of magnitude of τ - the robustness result of Section 3.

A.19 Lift and expected calibration

$$lift = AP / \pi, \quad ECE = \sum_b (|B_b|/n) |acc(B_b) - conf(B_b)| \quad (A24)$$

Lift normalizes AUPRC for prevalence, which differs between Gerdes (17.2%) and the PEC subset, making cross-yardstick comparisons meaningful.